

EXP 22: Prolog Program to Check if a Bird Can Fly

Name: Y. Roshitha (192424400)

Aim:

To develop a Prolog program that determines whether a particular bird can fly or not, and to verify the result using required queries.

Algorithm:

- Step 1: Start the program.
- Step 2: Create a database containing different birds.
- Step 3: Identify birds that are unable to fly and store them separately.
- Step 4: Select a bird from the database.
- Step 5: Check whether the selected bird belongs to the non-flying category.
- Step 6: If the bird is not in the non-flying list, conclude that it can fly.
- Step 7: Otherwise, conclude that it cannot fly.
- Step 8: Display the result for all birds or for a specific query.
- Step 9: Stop the program.

Source Code:

```
% Program to check whether a particular bird can fly or not
```

```
bird(sparrow).
```

```
bird(eagle).
```

```
bird(penguin).
```

```
bird(ostrich).
```

```
bird(parrot).
```

```
cannot_fly(penguin).
```

```
cannot_fly(ostrich).
```

```
can_fly(X) :-
```

```
    bird(X),
```

```
    \+ cannot_fly(X).
```

```
check_all :-
```

```
    forall(bird(X),
```

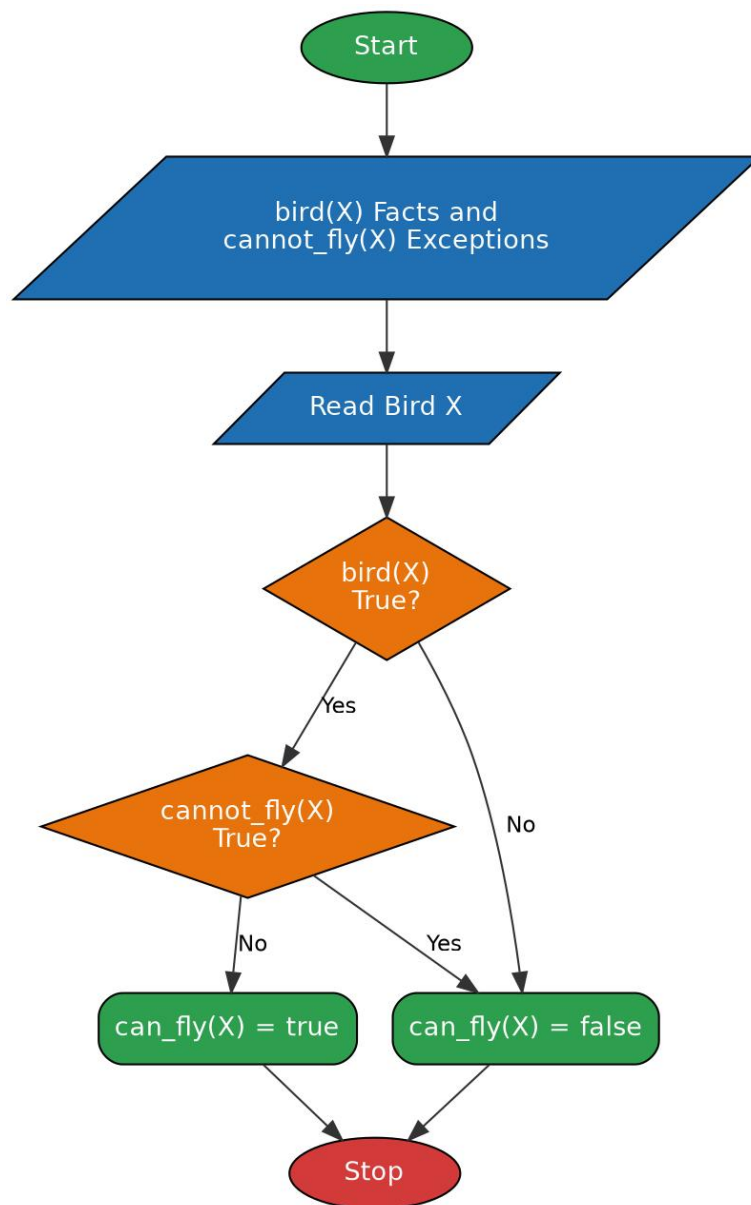
```
        ( can_fly(X)
```

```
        -> format("~w can fly.~n", [X])
```

```
        ; format("~w cannot fly.~n", [X])
```

```
        )).
```

Flowchart:



Output:

```
Welcome to SWI-Prolog (threaded, 64 bits, version 9.2.9)
SWI-Prolog comes with ABSOLUTELY NO WARRANTY. This is free software.
Please run ?- license. for legal details.
```

```
For online help and background, visit https://www.swi-prolog.org
For built-in help, use ?- help(Topic). or ?- apropos(Word).
```

```
?- cd('C:/Users/yanar/Documents/CSA1717').
true.
```

```
?- [birds].
true.
```

```
?- check_all.
sparrow can fly.
eagle can fly.
penguin cannot fly.
ostrich cannot fly.
parrot can fly.
true.
```

```
?- can_fly(eagle).
true.
```

```
?- can_fly(penguin).
false.
```

Result:

Thus the Prolog program to check whether a particular bird can fly or not was written, executed successfully and the output was verified.